

6:Future Enhancements (Performance Testing Roadmap)

Purpose

To outline upcoming improvements and expansion areas in the IDP Login performance testing strategy. This ensures alignment between QA, DevOps, and leadership on the roadmap and priorities.

1. Endurance Runs (8–10 Hours)

- **Current State:** Most runs are baseline (10–60 mins). Occasional 2–4 hr runs.
- **Gap:** Deadlocks and memory leaks typically appear after long-duration steady load.
- **Enhancement Plan:**
 - Schedule **8–10 hr endurance runs** at least once per release cycle.
 - Run at **50 VU or higher sustained load**.
 - Monitor DB locks, GC behavior, CPU/memory usage, and session stability.
 - Capture long-term trend data for throughput vs. latency.
- **Benefit:** Detect stability issues (deadlocks, resource leaks) before they reach production.

2. MFA Performance Testing

- **Current State:** Login scripts require MFA disabled for test accounts. MFA flows are traffic-only (not validated).
- **Gap:** Real-world users often use MFA missing in current SLA coverage.
- **Enhancement Plan:**
 - Develop **MFA-enabled login scripts** (covering PIN request, SMS/email flows).
 - Include **MFA page response times** in SLA.
 - Parameterize MFA data handling in CSVs.
- **Benefit:** Closer alignment with **production-like login scenarios**, improved risk coverage.

3. Adding Universal Services

- **Current State:** Current focus is IDP login, auth-server, metadata, enrollment, withdrawal.
- **Gap:** Broader "Universal Services" (e.g., cross-service APIs, participant services) not yet performance validated.
- **Enhancement Plan:**
 - Extend JMeter scripts for **Universal Enrollment, Plan, Participant flows**.
 - Add **composite scenarios** (login enrollment metadata) to simulate realistic user journeys.
 - Capture cross-service SLA impact.
- **Benefit:** Broader system coverage ensures end-to-end flows meet performance expectations.

4. CI/CD Nightly Integration

- **Current State:** Jenkins runs are manual/ad-hoc, no automated gating on SLA.
- **Gap:** No continuous visibility of regressions in nightly builds.
- **Enhancement Plan:**
 - Integrate performance suite into **CI/CD nightly jobs**.
 - Automate pass/fail against SLA thresholds (Child Page 1).
 - Push **daily summary email** with SLA compliance results.
 - Retain last 7 days of BlazeMeter reports for trending.
- **Benefit:** Early detection of regressions, **shift-left performance testing**, leadership-ready daily metrics.

5. Long-Term Considerations

- **Performance Environment Setup** – Dedicated infra for consistent results.
- **Service Virtualization** – Mock external dependencies to isolate bottlenecks.
- **Cloud Scaling with BlazeMeter** – Regional load generation for geo-distribution scenarios.
- **Automation of Report Archival** – Auto-publish results into Confluence dashboards.